

Definitions of Population Parameters:

1. Population Mean: $\mu = \frac{\sum_{i=1}^N x_i}{N}$ (Equals to average value)
2. Finite Population Variance: $\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$
Population standard deviation (SD): $\sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$
3. Covariance: $Cov(X, Y) = \frac{1}{N} \sum_{i=1}^N [(y_i - \mu_Y)(x_i - \mu_X)] = E(XY) - E(X)E(Y)$
4. Correlation: $\rho = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$
5. $U = \sum a_i X_i \rightarrow Var(U) = \sum_{i=1}^n a_i^2 Var(X_i) + \sum_{i=1}^n \sum_{j \neq i} a_i a_j Cov(X_i, X_j)$

Estimation:

we would like our estimation to be close to $\theta \Rightarrow |\hat{\theta} - \theta|$ to smaller than some number d . $|\hat{\theta} - \theta|$ is **error of estimation**

$Pr(|\hat{\theta} - \theta| < d) = 1 - \alpha$, where α is a small number and d is called **margin of error**.

样本量 n 决定估计量 $\hat{\theta}$ 的抽样分布特性

- n 很小 $\rightarrow$ 分布复杂且未知

- n 足够大 (相对于无限总体) $\rightarrow$ CLT 生效, 可用正态等熟悉分布近似

- n 很大且接近总体规模 $N \rightarrow$ 抽样方差急剧降低, 但需 FPC, 分布不再是简单的正态

If we are confident enough to say $\frac{\hat{\theta} - E[\hat{\theta}]}{\sigma_{\hat{\theta}}} \sim N(0, 1)$

Then we have $P(\hat{\theta} - \sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}} < E[\hat{\theta}] < \hat{\theta} + \sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}}) = 1 - \alpha$

If $\hat{\theta}$ is unbiased $\Rightarrow E[\hat{\theta}] = \theta$ then we have

$P(-\sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}} < \hat{\theta} - \theta < \sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}}) = 1 - \alpha \Rightarrow P(|\hat{\theta} - \theta| < \sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}}) = 1 - \alpha$ meaning we have found our sensible number, d

The interval above is the $100 \times (1 - \alpha)$ **confidence interval** for parameter

For a observation of $\hat{\theta}$ we want $E[\hat{\theta}] - \theta = 0$ (**bias**) and $E[(\hat{\theta} - \theta)^2]$ (**MSE**) is small

$MSE(\hat{\theta}) = Var(\hat{\theta}) + Bias(\hat{\theta})^2$, if estimate unbiased MSE equals its variance.

Three words describe an estimator: 1. **Unbiased** 2. **Precise** (Var is small) 3. **Accurate** (Unbiased + Precise)

Survey Terminology:

1. Population: A collection of elements about which we want to make inferences
2. Element: An element is an object on which a measurement is taken
3. Sampling Units: Partition the population
4. Frame: A frame is a list of sampling units
5. Sample: A collection of sampling units drawn from a single or from multiple frames

Probability Sampling: Each population unit has a known and non-zero probability of being selected, and the sampling process relies entirely on a random mechanism. **Can make probability statements (e.g. Confidence Intervals)** Randomisation balances out all factors inherent in a population

Quota Sampling: Quotas are first established based on certain characteristics (such as gender, age, region), then interviewers **subjectively** or conveniently select samples within each quota. **Can not make probability statements**

Simple Random Sampling(SRS)

Notation (抽样率 & f.p.c.)

- **population size** N , **sample size** n , **sampling fraction** $f = \frac{n}{N}$, **finite population correction** f.p.c. $= 1 - f = 1 - \frac{n}{N}$ 。
- **population mean** $\mu = \frac{1}{N} \sum_{j=1}^N u_j$ 。样本观测记为 $Y_1, \dots, Y_n$, 样本均值 $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$ 。
- 记 $Z_j = \mathbf{1}\{\text{单位 } j \text{ 被抽中}\}$, 则 $\bar{Y} = \frac{1}{n} \sum_{j=1}^N u_j Z_j$ 。

所有大小为 n 的样本等概率; 任意单位的包含概率: $\Pr(Z_j = 1) = \frac{n}{N}$ 。

Proof (含 $\Pr(Z_j = 1)$) $\Pr(Z_j = 1) = \frac{\binom{N-1}{n-1}}{\binom{N}{n}} = \frac{n}{N}$ 。

Estimating a mean μ

(A) **Unbiasedness** $\mathbb{E}[\bar{Y}] = \mu$ 。

Proof $\mathbb{E}[\bar{Y}] = \frac{1}{n} \sum_{j=1}^N u_j \mathbb{E}[Z_j] = \frac{1}{n} \sum_{j=1}^N u_j \frac{n}{N} = \mu$ 。

(B) **Variance of $\bar{Y}$**

$$\text{Var}(\bar{Y}) = \frac{1 - \frac{n}{N}}{n} \cdot \frac{1}{N-1} \sum_{j=1}^N (u_j - \mu)^2 = \frac{N-n}{N-1} \frac{\sigma^2}{n}。$$

Proof

记 $a_j = \frac{u_j}{n}$ 。有

$$\text{Var}(\bar{Y}) = \sum_j a_j^2 \text{Var}(Z_j) + \sum_{j \neq k} a_j a_k \text{Cov}(Z_j, Z_k)。$$

$$\text{SRS 下 } \text{Var}(Z_j) = \frac{n}{N} \left(1 - \frac{n}{N}\right), \quad E(Z_j) = \frac{n}{N}$$

$$\text{Cov}(Z_j, Z_k) = E(Z_j Z_k) - E(Z_j)E(Z_k) = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2}。$$

$$= -\frac{n}{N} \left(1 - \frac{n}{N}\right) \frac{1}{N-1}$$

(C) **Unbiased estimator of $\text{Var}(\bar{Y})$**

$$\widehat{\text{Var}}(\bar{Y}) = \frac{1 - \frac{n}{N}}{n} \cdot \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2。$$

Proof

From B we have $\text{Var}(\bar{Y})$ with σ^2 , however it is hard to know.

Noticed that if we use $s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$ to represent $\hat{\sigma}^2$. It is biased $E[\hat{\sigma}^2] = \frac{N}{N-1} \sigma^2$

Therefore, $(1 - \frac{1}{N})\hat{\sigma}^2$ is an unbiased estimator of σ^2 , then we have our $\widehat{\text{Var}}(\bar{Y})$

(D) t-CI for μ

$$\mu \in \bar{y} \pm t_{n-1, \alpha/2} \sqrt{\frac{1 - \frac{n}{N}}{n} \cdot \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2}.$$

Sample size for μ (目标半宽 d)

$$n = \frac{N z_{\alpha/2}^2 v^2}{d^2 (N-1) + z_{\alpha/2}^2 v^2}. \text{ 其中 } v \text{ 是方差, 如果不知道可以用 } s$$

$$\text{若用 } t \text{ 值, 解 } n = \frac{N t_{n-1, \alpha/2}^2 v^2}{d^2 (N-1) + t_{n-1, \alpha/2}^2 v^2}; \text{ 因 } t \text{ 随 } n \text{ 变化, 需迭代试算。}$$

Estimating a total $\tau = N\mu$

结论 (点估计 / 方差估计 / **t-CI**)

$$\hat{\tau} = N \bar{y}.$$

$$\text{Var}(\hat{\tau}) = \frac{N-n}{N-1} \frac{N^2}{n} \sigma^2$$

$$\widehat{\text{Var}}(\hat{\tau}) = N^2 \cdot \frac{1 - \frac{n}{N}}{n} \cdot \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2.$$

$$\tau \in N \bar{y} \pm t_{n-1, \alpha/2} N \sqrt{\frac{1 - \frac{n}{N}}{n} \cdot \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2}.$$

样本量 (目标半宽 d)

$$n = \frac{N z_{\alpha/2}^2 v^2}{\left(\frac{d}{N}\right)^2 (N-1) + z_{\alpha/2}^2 v^2}.$$

Estimating a proportion p

把 0/1 变量视作均值问题: 令 $x_i \in \{0, 1\}$, 样本比例 $\hat{p} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$.

$$\text{Var}(\hat{p}) = \frac{N-n}{N-1} \frac{1}{n} p(1-p)$$

$$\widehat{\text{Var}}(\hat{p}) = \frac{1 - \frac{n}{N}}{n} \hat{p}(1-\hat{p})$$

$$p \in \hat{p} \pm t_{n-1,\alpha/2} \sqrt{\frac{1 - \frac{n}{N}}{n} \hat{p}(1 - \hat{p})}$$

(常用条件: $n\hat{p} \geq 5, n(1 - \hat{p}) \geq 5$)